

Andy Mannikum

✉ andymannikum76@gmail.com ☎ +447765535609 📍 Glasgow

🌐 <https://andym764.github.io/>  [linkedin.com/in/andy-mannikum-21a84b3b5](https://www.linkedin.com/in/andy-mannikum-21a84b3b5)

Profile

Mechatronics Engineering student at the University of Glasgow with hands-on experience in control systems, signal processing, and embedded systems. Recently completed a research internship in Japan focused on neural signal analysis and hardware–software integration for closed-loop stimulation systems. Strong background in MATLAB/Simulink, C/C++, and industrial automation, with a particular interest in robotics, intelligent control, and data-driven engineering systems.

Education

University of Glasgow <i>BEng Mechatronics</i>	09/2022 – 05/2027
Royal College of Port Louis <i>Cambridge International Examinations (CIE) - Higher School Certificate</i> Main Subjects: Physics (A), Design and communication (A), Mathematics (A*)	01/2021 – 11/2021

Professional Experience

University of Sheffield <i>Project Assistant</i> - Contributing to the development of an AI-powered smart tennis training system integrating computer vision, motion tracking, and robotics technologies. - Developing software for Raspberry Pi-based camera systems to support real-time tennis ball and player tracking. - Working with computer vision and TrackNet-inspired techniques for ball trajectory detection and analysis. - Developing mobile application interfaces for wireless communication with embedded hardware devices. - Supporting the integration of sensor data, video analytics, and machine learning methods for automated performance assessment. - Collaborating on the development of intelligent tennis ball machine technologies aimed at enhancing player training and feedback.	06/2026 – 07/2026 Sheffield, Uk
INOPASE <i>Engineering Intern</i> - Developed MATLAB-based signal processing pipelines to analyse neural activity and identify correlations between stimulation inputs and physiological responses - Applied filtering and data analysis techniques to extract meaningful features from experimental rat study datasets - Investigated concepts of closed-loop control for adaptive neural stimulation systems	02/2026 – 04/2026 Tokyo, Japan

- Interfaced with microcontroller-based PCB systems for data acquisition and experimental testing
- Supported hardware–software integration, debugging, and optimisation of embedded systems
- Reviewed and applied insights from scientific literature to support experimental design and data interpretation

RT Knits Ltd

08/2025 – 09/2025

Industrial Automation Intern

Mauritius

- Assisted in the design, integration, and troubleshooting of automated production systems in a textile manufacturing environment
- Supported implementation of PLC-controlled systems, sensors, and actuators to improve efficiency and reduce downtime
- Conducted preventive maintenance and calibration of automation equipment
- Analysed production workflows and proposed improvements to enhance throughput and quality

Sparrow Distribution Co Ltd

06/2024 – 08/2024

Industrial Automation Intern

Mauritius

- Assisted in operation and troubleshooting of automated packaging and robotic systems
- Gained hands-on experience with conveyor systems, sensors, and industrial automation equipment
- Supported machine setup, calibration, and preventive maintenance to minimise downtime
- Contributed to workflow optimisation and quality control processes

Technical Interests

- Control Systems & Feedback Control
- Signal Processing & Data Analysis
- Embedded Systems & Microcontrollers
- Robotics & Automation

Skills

Programming & Tools

- Python
- C/C++
- MATLAB & Simulink
- Linux
- ROS2

Engineering Tools

- Fusion 360 (CAD)
- AutoCAD
- PLC Systems & Industrial Automation

Projects

Autonomous Vegetation Monitoring Robot

09/2026

Final Year Project

- Designing an autonomous wheeled mobile robot for vegetation monitoring and inspection across row-based environments.
- Developing a model- and simulation-driven design workflow using MATLAB/Simulink and Gazebo, covering requirements capture, robot modelling, navigation and system evaluation.

- Integrating ROS2, Raspberry Pi, sensors and computer vision to enable autonomous navigation, plant detection and inspection.
- Developing closed-loop motor control and guidance/navigation for reliable movement through vegetation rows.
- Developing a mobile application interface to communicate with the Raspberry Pi and display robot status, inspection data and plant-health results.
- Evaluating the physical prototype against simulation through experimental testing and iterative design improvements.

Flight Control System

01/2025 – 04/2025

Year 3

Collaborated in a multidisciplinary team to design and implement a flight control system for a restricted-motion twin-rotor UAV. Developed a mathematical model of the UAV dynamics and validated it using experimental test data. Designed and simulated PID controllers for the azimuth, pitch, and elevation subsystems to achieve stable and accurate control. Integrated the control algorithms with the hardware test rig and successfully demonstrated controlled UAV operation. This project provided practical experience in control system design, dynamic system modelling, simulation, teamwork, and technical presentation.

Instrument Landing System (ILS) lateral beam guidance model

09/2024 – 12/2024

Year 3

As part of a third-year Simulation of Engineering Systems module at the University of Glasgow, I developed a MATLAB and Simulink simulation of an aircraft Instrument Landing System (ILS) lateral beam guidance model. The system simulated aircraft alignment with the runway centreline and analysed behaviours such as beam capture, lateral deviation, and glide path tracking. The model was also used to evaluate system performance under disturbances such as crosswinds, providing practical experience in dynamic system modelling and control system simulation.

Project Freeheart

01/2024 – 03/2024

Year 2

Built an embedded heart rate monitor based on Photoplethysmography. The device was developed with Mbed OS (C++) on an NXP KL25Z microcontroller. Additionally, Python was employed to model and simulate the processing of data samples on a PC.

Low Step Bike Mechanical Design

01/2024 – 03/2024

Year 2

Designed a low-step bicycle using Fusion 360 to improve accessibility and rider comfort by enabling easier mounting and dismounting. Developed the frame geometry using CAD tools while considering structural stability and ergonomic design. Performed basic stress and force analysis within the software to evaluate load distribution and ensure the structural integrity of the frame.

Model Lift (Elevator)

09/2022 – 12/2022

Year 1

Worked as part of a team to design and implement a fully functional model elevator system programmed in C++. Developed control logic to manage lift movement, floor sequencing, and operational data handling. Enhanced the system by implementing a real-time LED floor display to indicate lift position and improve user interaction. Integrated switches and indicator circuits to simulate the behaviour of a realistic elevator control system.

Interests

Regularly play football with friends and exercise to keep fit |

Passionate about cooking and exploring new dishes |

Enthusiastic hiker with a love for outdoor adventures

Languages

English

- Fluent, with strong oral and written communication skills developed through professional and academic experiences

French

- Fluent, with comprehensive ability to read, write, and speak, supported by extensive exposure to French- speaking environments

Hindi

- Fluent, with comprehensive ability to read, write, and speak